

Neural Networks

06 — Bayesian Inference & Uncertainty CosmicAI Summer School 2026

Your model has an answer. But does it know what it doesn't know?

Every fit so far has produced a **point estimate** — a single best value.

Warning

A point estimate is not the whole story.

- How confident is the fit?
- What if the data are scarce, or the model is uncertain?
- How do we encode prior knowledge from physics?

Today: two frameworks for reasoning under uncertainty, their relationship to model fitting, and how they connect to neural networks.

Sampling as integration — Monte Carlo

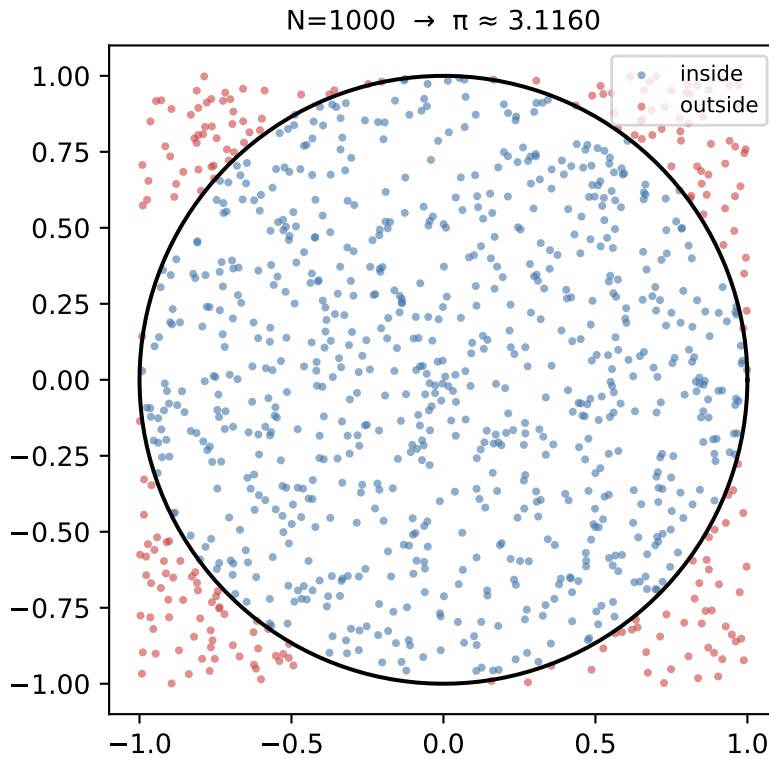
Many integrals are hard analytically — easy to *approximate by sampling*.

$$\hat{\pi} = 4 \times \frac{\text{inside}}{\text{total}}$$

Converges as $1/\sqrt{N}$ — halving the error costs $4\times$ the samples.

This is the foundation of MCMC — the engine of Bayesian inference. MCMC draws *correlated* samples from a distribution you cannot write analytically.

→ *Notebook 06, Part 1*



Two interpretations of probability

The mathematics of probability is the same for both camps. The **meaning** is different.

	Frequentist	Bayesian
Probability means	Long-run frequency	Degree of belief
Parameters are	Fixed unknowns	Random variables
You get	Best-fit + confidence interval	Posterior distribution
Prior knowledge	Not encoded formally	Encoded as $P(\theta)$
Core question	$P(\text{data} \mid \cdot)$	$P(\cdot \mid \text{data})$
Key tool	MLE, χ^2 fitting	Bayes' theorem + MCMC

i Note

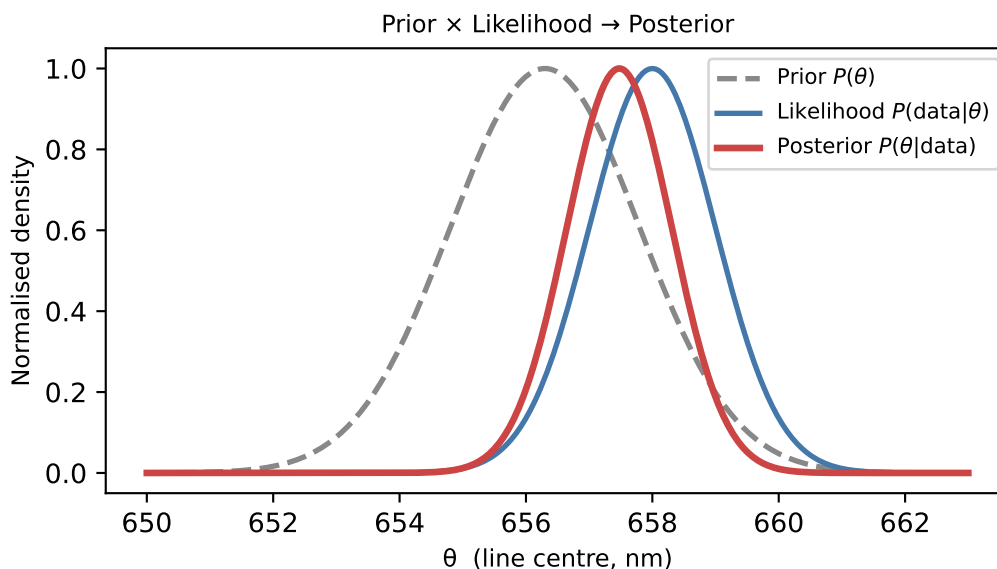
A confidence interval is **not** a probability statement about the parameter. A Bayesian credible interval **is**.

Bayes' theorem

$$P(\theta \mid \text{data}) = \frac{P(\text{data} \mid \theta) P(\theta)}{P(\text{data})}$$

- $P(\theta \mid \text{data})$ — posterior: belief after data
- $P(\text{data} \mid \theta)$ — likelihood: how probable is this data if θ is true?
- $P(\theta)$ — prior: belief before data
- $P(\text{data})$ — normalising constant, usually ignored

Under Gaussian noise: minimise χ^2 — maximise log-likelihood.



Model fitting: the unified view

Both approaches share the same **model** and the same **likelihood**. The difference is what you do with the prior.

The model: a Gaussian emission line on a flat continuum

$$\text{flux}(\lambda) = 1 + A \cdot \exp\left(-\frac{(\lambda - \lambda_0)^2}{2\sigma^2}\right) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_n^2)$$

Free parameters: A (amplitude), λ_0 (line centre in nm).

Prior from physics: $\lambda_0 \sim \mathcal{N}(656.3, 1.0^2)$ — H, Doppler-shifted by at most ± 1 nm.

	Frequentist	Bayesian
Prior on λ_0	Not used	$\mathcal{N}(656.3, 1.0^2)$
Result	Best-fit + covariance	Posterior $P(A, \lambda_0 \mid \text{data})$
Marginalisation	Ad hoc	Native

When Bayesian wins: the faint detection

At low SNR, the likelihood surface is broad. The prior anchors the posterior where physics says the line must be.

Reading a corner plot:

- *Diagonal*: marginal posterior for each parameter
- *Off-diagonal*: joint posterior (2D histogram)
- *Dashed lines*: 16th, 50th, 84th percentiles
- *Red star*: true values
- *Tilt in the 2D panel*: correlation between A and λ_0

At $\text{SNR} = 2$, the posterior is wide. The prior on λ_0 prevents the chain from wandering to unphysical values.

Note

What MLE cannot do here:

- MLE returns one point. It cannot represent multi-modal posteriors.
- MLE has no native way to marginalise over nuisance parameters.
- At $\text{SNR} = 2$, MLE may converge on a noise spike.

The Bayesian posterior shows you the *full* uncertainty — including correlations.

→ *Notebook 06, Part 3 — change SNR with the dropdown*

When Frequentist wins: large N

At high SNR / large N , the likelihood dominates. The prior stops mattering.

Three reasons to use MLE when N is large:

1. **Bernstein–von Mises theorem:** As $N \rightarrow \infty$, the posterior concentrates around the MLE and becomes Gaussian with covariance equal to the Fisher information. Your prior is irrelevant — any prior gives the same answer.

2. **Cramér–Rao efficiency:** MLE is asymptotically unbiased and achieves the *minimum possible variance* for any unbiased estimator. You cannot do better.
3. **Computational cost:** `scipy.optimize.curve_fit`: milliseconds. MCMC with 32 walkers $\times$ 2000 steps: minutes. DESI is measuring 40 million spectra — only MLE is practical.

i Note

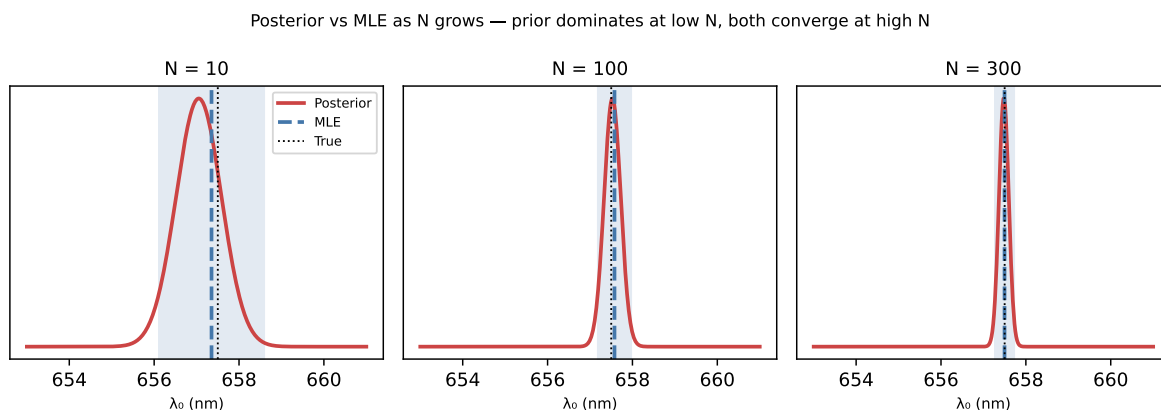
They **converge to the same answer**.

The frequentist path is faster and provably optimal when data are abundant.

The Bayesian prior buys you nothing — and costs computation — when N is large.

→ *Notebook 06, Part 4* — move the N slider

The convergence in pictures



When to use each

Use Bayesian inference when:

- Data are scarce (low N or low SNR)
- You have genuine prior knowledge that should constrain the fit
- You need a full posterior (marginalisation, propagation of uncertainty)
- The posterior is multi-modal and MLE would miss solutions

Use MLE when:

- Data are abundant and the likelihood is well-behaved
- You need speed (survey pipelines, parameter grids)

- The prior is vague and would be arbitrary anyway

Neural networks and uncertainty

A trained neural network produces a **point estimate** — no posterior, no uncertainty.

Bayesian neural networks (BNNs):

Instead of fixed weights w , treat weights as random variables with a posterior:

$$P(w \mid \text{data}) \propto P(\text{data} \mid w) P(w)$$

Prediction is marginalised over the weight posterior:

$$P(y \mid x, \text{data}) = \int P(y \mid x, w) P(w \mid \text{data}) dw$$

This integral is intractable for large networks — we need approximations.

Types of uncertainty:

- Epistemic: uncertainty from limited data — *reducible* with more data
- Aleatoric: irreducible noise in the data itself

A network that does not model uncertainty conflates the two. It can be confidently wrong in regions far from training data.

MC Dropout: approximate Bayesian inference

Gal & Ghahramani (2016): dropout active **at test time** Bayesian inference.

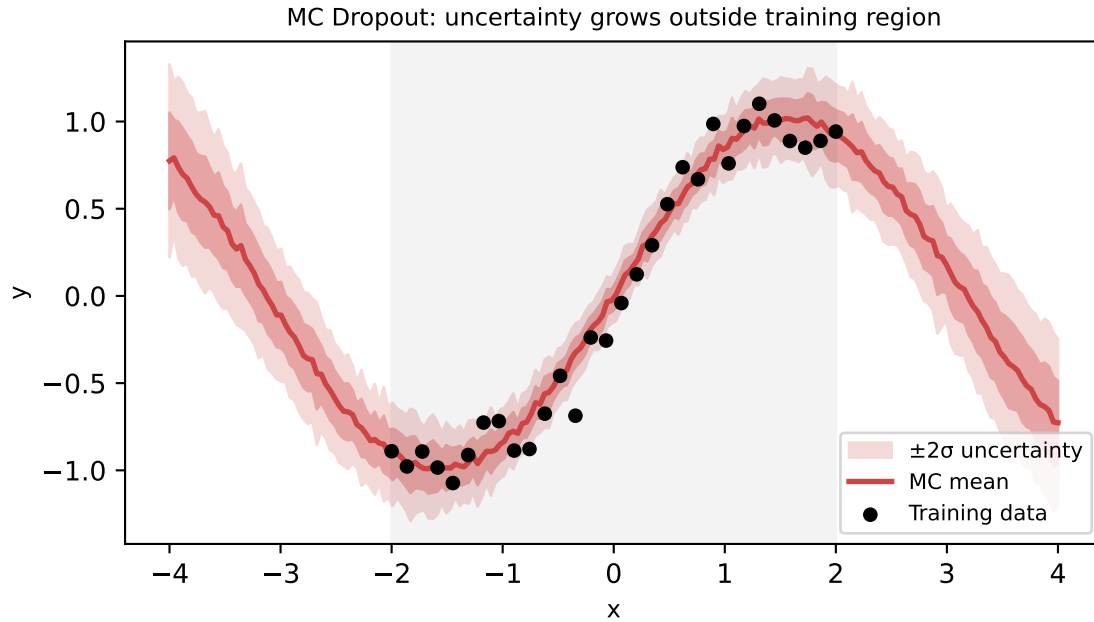
Each mask samples the weight posterior. N passes give:

- **Mean:** predictive mean
- **Std:** epistemic uncertainty

$$\text{uncertainty}(x) \approx \text{Std}[\{f_w(x)\}_{i=1}^N]$$

Uncertainty grows far from training data — the network knows what it doesn't know.

→ *Notebook 06, Part 5*



When the likelihood is intractable

Everything above assumed you can write $P(\text{data} \mid \theta)$. For many problems in astronomy, you cannot.

Examples:

- **Gravitational waves:** signal model is a full waveform simulation
- **Cosmological inference:** connecting Ω_m, σ_8 to clustering requires N-body simulation
- **Stellar population synthesis:** a single SED marginalises over millions of stellar tracks

The problem:

MCMC requires evaluating $P(\text{data} \mid \theta)$ at every step.

If one evaluation costs minutes of simulation time — MCMC is dead.

We need a different approach.

Simulation-Based Inference

SBI replaces the likelihood with a learned surrogate:

$$(\theta_{\text{sim}}, x_{\text{sim}}) \xrightarrow{\text{neural posterior estimator}} \hat{P}(\theta \mid x_{\text{obs}})$$

1. Simulate millions of (parameter, observation) pairs
2. Train a normalising flow to approximate the posterior directly
3. Evaluate on real observations — no likelihood required

The neural network *is* the inference engine.

i Note

Cranmer, Brehmer & Louppe (2020) The frontier of simulation-based inference. PNAS 117(48).

References

Papers

- Gal & Ghahramani (2016) — *Dropout as a Bayesian Approximation*. ICML. arXiv:1506.02142
- Cranmer, Brehmer & Louppe (2020) — *The frontier of SBI*. PNAS 117(48). arXiv:1911.01429
- Foreman-Mackey et al. (2013) — *emcee: The MCMC Hammer*. PASP 125(925). arXiv:1202.3665
- Foreman-Mackey (2016) — *corner.py*. JOSS 1(2)

Books & resources

- Gelman et al. — *Bayesian Data Analysis* (3rd ed.) — free PDF at stat.columbia.edu/~gelman/book